

## **Beyond the Ballot Box:**

### **Micro-Donation Patterns, Voter Turnout, and Political Engagement Typologies Across the United States**

#### **Executive Summary**

Political campaigns are constrained by two fundamental scarce resources: time and money. For any campaign to succeed, strong funding is needed to effectively communicate a platform and secure broad support. In a modern landscape where data-driven insights like targeted advertising, tailoring digital feeds, and predictive modeling are used to optimize resource allocation, it is logical that these same tools should be used to guide political candidates to tighten their operations. Political contributions are significant for primarily funding campaign operations, but they can also serve as a metric of grassroots momentum. Understanding the intersection of donors and actual voters is important to build a sustainable support base and to expand a candidate's reach. This study aims to use data from the 2024 election cycle in the U.S. to identify demographic and geographic drivers of small-dollar political contributions to optimize outreach.

The analysis integrated four primary datasets to construct a robust model of political engagement. Financial data was sourced from the Federal Election Commission (FEC), encompassing 2023–2024 individual contribution records and committee master files. These records provided donor-specific details, including geographic ZIP codes and contribution amounts. To isolate grassroots behavior, the sample was filtered to micro-donations of \$200 or less, resulting in a dataset comprising approximately 50 million observations.

Socioeconomic metrics from the 2024 U.S. Census Bureau were extracted at the ZIP code level, providing eight features such as educational attainment, median household income, and homeownership rates. Using data from the U.S. Department of Urban Planning and Development, the zip-code data were reallocated into county aggregates, using population weighting to prevent statistical distortion due to counties that spanned multiple zip codes. Lastly, to contextualize donation patterns within electoral outcomes, county-level voter turnout data for the 2024 election were integrated from the MIT Election Data and Science Lab (MEDSL).

With the consolidated data, a multi-stage analytical pipeline was run to identify the primary features driving political donations. First, a Random Forest model was used to uncover feature importance. This specific model was selected for its ability to capture non-linear relationships and handle the multicollinearity expected between variables such as education and income. While this provided a foundational understanding of the data, the Random Forest's complexity necessitated further modeling for causal inference.

Therefore, a Best Subset Selection OLS Regression was performed on two separate dependent variables: micro-donation volume for each major party. This process tested all 255 possible variable combinations and was optimized using the Bayesian Information Criterion (BIC) to ensure the resulting models were not overfit and were statistically significant. Using these

findings as a guide, GIS software was used to visualize these patterns geographically. Finally, a K-Means clustering algorithm grouped all 3,097 counties into four discrete profiles, which can be visualized as a 2x2 matrix based on high/low voter turnout and high/low donor activity.

The results of these models identified distinct national and partisan drivers of political engagement. Nationally, the Random Forest model revealed that educational attainment is the premier predictor of micro-donation volume, with a relative importance score nearly double that of the next most significant variable, median household income. The OLS models further highlighted fundamental partisan differences: for Democrats, education remains the strongest positive driver, yet higher median income correlates with a decrease in donation likelihood when controlled for other factors. Conversely, Republican micro-donations are primarily driven by homeownership rates. Despite these diverging demographic drivers, both parties share median age as a significant positive predictor, confirming that older populations remain the most consistent source of small-dollar financial support across the American electorate.

To maximize limited campaign resources, these findings provide a framework for a tailored outreach strategy based on the four distinct engagement clusters identified. In High-Turnout / Low-Donation zones, characterized primarily by older homeowners, campaigns should focus on "donor conversion" by building trust through local community issues like property tax policy and senior services, to translate existing political excitement into financial funding. In High-Donation / Low-Turnout areas, the priority must shift from persuasion to mobilization to ensure that high levels of financial commitment translate into ballots at the polls. For High-Turnout / High-Donation areas, which often exhibit lower income and education levels but high engagement, outreach should pivot toward populist messaging that frames micro-donations as an investment in gaining a voice. Finally, in Low-Turnout / Low-Donation areas that coincide with high income and education, candidates must overcome this population's perceived barriers for engagement. The most likely solution would be to present a professional campaign with clear and well-thought-out policies and a path to victory to prove that participation is worth the voters' time. These typologies serve as essential navigational markers: Democratic candidates should prioritize older, educated voters, while Republican efforts have the highest returns among older, established homeowner populations in less densely populated regions.